

# Rave Generation

PRESENTS

## RAVE STATION 2



USER MANUAL

# Introduction

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RaveStation 2 draws inspiration from iconic workstations, samplers, and effects of the mid-90s. Featuring a 16-voice polyphonic engine with filter modeling and modulation, it offers a vast array of sounds, effect and synthesis parameters designed to capture the essence of electronic music production from this era. This makes it ideal for a wide range of electronic dance music genres, particularly excelling in styles like Rave, Techno, mid-90s Hardcore, Eurodance, House, and Old School.

## 900+ Sounds / 2,000+ Samples

RaveStation 2 includes 900+ sounds built from 2,000+ samples (40 minutes of audio), all tuned to key with multi-sample support. Sounds are organized into the following categories:

- BASS - Deep basses, sub basses, and bass leads
- FX - Risers, impacts, sweeps, and special effects
- HOOVER - Classic rave hoovers and reeses
- MULTI - Breakbeat, Vocal, and Drum kits
- ORGAN - Organs and keyboard sounds
- STAB 1-3 - Classic stabs and chord hits
- DRUMS - Single kicks, snares, toms, hats and percussion
- SYNTH 1-3 - Leads, pads, plucks, and synthesizer sounds

## Key Features

- 16-voice polyphony with intelligent voice allocation
- 11 filter types including Roland, Moog, Juno and TB-303 emulations
- Three ADSR envelopes (Amp, Filter, Pitch)
- LFO with 8 waveforms, 9 destinations, and tempo sync
- JP-8000 Supersaw unison and symmetric detune
- 14-pattern SH-101 style arpeggiator
- Chorus (6 modes) with pre-delay, Ping-Pong/Stereo Delay, and Reverb with ducking
- Stereo width control, Bass/Mid/Treble tone shaping and OTT multiband dynamics
- Full preset management via toolbar browser

## User Interface Tips

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RaveStation 2 uses a streamlined interface where sounds are selected via the preset browser in the menu bar. The main panel provides direct access to all synthesis and effect parameters.

### Tooltips

Hover a switch to see what it does.

### Precise Movements

Holding down the Shift key while adjusting a knob will enable you to make precise movements. This allows for finer control over the audio processing parameters.

### Reset Knob

Right-clicking a knob will reset its value. This action restores the knob or fader to its default or initial position. Alt+click also works (macOS: Option+click).

### Adjust Value

Double-clicking a knob allows you to adjust its value directly by typing. This provides a quick and intuitive way to fine-tune parameters with precision.

### Undo/Redo

Access Undo and Redo from the menu bar to easily correct or reapply changes. Undo reverses the last adjustment, while Redo restores it.

### GUI Opacity

From the menu bar, you can adjust the GUI Opacity from 0% to 100%. This control allows for varying the transparency of the plugin interface, offering better visibility of underlying work areas.

### Resize

The interface can be resized from 70% to 200% of its original size via the menu bar. This feature accommodates different screen sizes and resolutions.

### Preset Menu

Accessible from the menu bar, the Preset Menu provides a comprehensive interface for browsing, loading, and selecting presets. Users can also Load/Save and Copy/Paste presets.

## Filter Section

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RaveStation 2 features a comprehensive filter section with 11 distinct filter types, each offering unique tonal characteristics. The filter can be modulated by the Filter Envelope, LFO, velocity, and key track.

### Filter Types

#### **LP CLEAN (12 DB/OCT)**

A transparent 2-pole lowpass filter suitable for gentle frequency shaping without coloration.

#### **LP ROLAND 24 (24 DB/OCT)**

Inspired by the Roland IR3109 filter chip. Two cascaded 12 dB stages create a warm, punchy 24 dB/oct response with smooth resonance.

#### **LP MOOG 24 (24 DB/OCT)**

Transistor ladder filter emulation with the classic Moog sound. Features nonlinear feedback for rich harmonics and musical self-oscillation at high resonance.

#### **HP SVF (12 DB/OCT)**

State Variable highpass filter. Excellent for removing low-end mud or creating thin, cutting textures.

#### **BP SVF (12 DB/OCT)**

State Variable bandpass filter. Perfect for telephone effects, wah-like sweeps, or isolating frequency ranges.

#### **PEAK**

Resonant peak filter that boosts a narrow frequency band. Great for adding presence or creating formant-like effects.

#### **NOTCH**

Rejects a narrow frequency band while passing others. Useful for removing unwanted frequencies or phaser-like effects.

#### **LP FM 12 (12 DB/OCT)**

Lowpass filter with audio-rate frequency modulation. The Drive control affects FM depth, creating dynamic harmonic content that responds to the input signal.

#### **LP 303 (24 DB/OCT)**

Authentic TB-303 diode ladder filter emulation. Features the distinctive 303 squelch and acidic resonance character with 150 Hz feedback highpass.

## LP JU6 24 (24 DB/OCT)

Four-pole OTA cascade with per-stage saturation and 4x internal oversampling, using the Juno-6 exponential resonance curve. The natural volume drop at high resonance comes from the feedback compression itself, as on the hardware.

## LP JU106 24 (24 DB/OCT)

Juno-106 voicing of the same OTA cascade, with the 106's polynomial resonance fit. Slightly softer resonance behaviour than the Ju6.

## Filter Parameters

| Parameter                 | Range         | Default  | Description                                              |
|---------------------------|---------------|----------|----------------------------------------------------------|
| Freq (Cutoff)             | 0.000-1.000   | 1.000    | Filter cutoff frequency (20 Hz to 20 kHz exponential)    |
| Reso (Resonance)          | 0.000-0.990   | 0.000    | Filter resonance/Q. Higher values create peaks at cutoff |
| HPF (HPF Cutoff)          | 0-100%        | 0%       | Highpass filter cutoff (8 Hz to 5 kHz). Jupiter-8 style  |
| Type (Filter Type)        | 11 types      | LP Clean | Select filter algorithm (see descriptions above)         |
| Drive                     | 0-100%        | 0%       | Filter saturation/overdrive amount                       |
| Vel Follow (VCF Velocity) | -100 to +100% | 0%       | Velocity modulation of filter cutoff                     |
| Drive Position            | Post/Pre      | Post     | Apply drive before or after filter                       |
| Key Track                 | -100 to +100% | 0%       | Filter tracking of keyboard pitch (Roland style)         |

## Amp Envelope

The Amp Envelope controls the volume contour of each voice from note-on to note-off and beyond. It shapes how the sound fades in, sustains, and releases.

## Parameters

| Parameter | Range  | Default | Description                                               |
|-----------|--------|---------|-----------------------------------------------------------|
| Attack    | 0-1000 | 0       | Time for volume to rise from zero to full (1ms to 2s)     |
| Decay     | 0-1000 | 1000    | Time to fall from full level to sustain level (1ms to 2s) |
| Sustain   | 0-100% | 100%    | Volume level held while key is pressed                    |
| Release   | 0-1000 | 0       | Time for volume to fade after key release (1ms to 5s)     |

## Filter Envelope

The Filter Envelope modulates the filter cutoff frequency over time, adding movement and expression to the sound. The envelope amount can be positive or negative.

## Parameters

| Parameter  | Range          | Default | Description                                             |
|------------|----------------|---------|---------------------------------------------------------|
| Attack     | 0-1000         | 0       | Time for envelope to rise to peak (1ms to 2s)           |
| Decay      | 0-1000         | 100     | Time to fall from peak to sustain level (1ms to 2s)     |
| Sustain    | 0-100%         | 0%      | Envelope level held while key is pressed                |
| Release    | 0-1000         | 50      | Time for envelope to fall after key release (1ms to 5s) |
| Env Amount | -1.00 to +1.00 | 0.00    | Depth and direction of filter modulation                |

## Pitch Envelope

The Pitch Envelope provides time-varying pitch modulation for each voice. Use it for pitch sweeps, attack transients, or subtle pitch drift effects.

## Parameters

| Parameter   | Range         | Default | Description                                             |
|-------------|---------------|---------|---------------------------------------------------------|
| Attack      | 0-1000        | 0       | Time for envelope to rise to peak (1ms to 2s)           |
| Decay       | 0-1000        | 100     | Time to fall from peak to sustain level (1ms to 2s)     |
| Sustain     | 0-100%        | 0%      | Envelope level held while key is pressed                |
| Release     | 0-1000        | 50      | Time for envelope to fall after key release (1ms to 5s) |
| Pitch Depth | -48 to +48 st | 0 st    | Maximum pitch deviation in semitones                    |

# LFO

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The Low Frequency Oscillator provides cyclic modulation for creating vibrato, tremolo, filter wobbles, and other time-varying effects. It supports tempo sync and fade-in/fade-out.

## Waveforms

- Sine - Smooth, rounded modulation
- Triangle - Linear rise and fall
- Square - Abrupt on/off switching
- Saw Up - Rising ramp
- Saw Down - Falling ramp
- Random - Smoothly interpolated random values
- S&H - Sample & Hold (stepped random)
- Triple Peak - Nord Lead 3 style (fundamental + 3rd harmonic)

## Destinations

- Off - LFO disabled
- Filter - Modulates filter cutoff (wah/wobble)
- Pitch - Modulates pitch (vibrato)
- Pan - Modulates stereo position (auto-pan)
- Amp - Modulates volume (tremolo)
- HPF - Modulates highpass filter cutoff
- Resonance - Modulates filter resonance
- Drive - Modulates filter drive amount
- Mid Freq - Modulates mid filter frequency

## Parameters

| Parameter                | Range       | Default | Description                           |
|--------------------------|-------------|---------|---------------------------------------|
| Type (Waveform)          | 8 types     | Sine    | LFO wave shape                        |
| LFO Dest (Destination)   | 9 targets   | Off     | Modulation target                     |
| Rate                     | 0.01-106 Hz | 2 Hz    | LFO speed (when not synced)           |
| LFO Sync (Sync Division) | 24 values   | 1/1     | Tempo-synced rate division            |
| Depth                    | 0-100%      | 50%     | Modulation intensity                  |
| Fade                     | -5 to +5 s  | 0 s     | Positive: fade in. Negative: fade out |

| Parameter              | Range    | Default | Description                                                                        |
|------------------------|----------|---------|------------------------------------------------------------------------------------|
| Retrig (LFO Retrigger) | Off / On | Off     | Restarts the LFO at 0 degrees on every note-on, for a consistent attack character. |

## Unison & Detune

RaveStation 2 features two independent systems for creating thick, wide sounds: Supersaw-style Unison and Symmetric Detune.

### Unison (Supersaw)

When set to 7 voices, it uses the authentic JP-8000 frequency ratios and detune curve. Other voice counts (2-6, 8-9) use a symmetric Hypersaw algorithm.

| Parameter | Range  | Default | Description                                    |
|-----------|--------|---------|------------------------------------------------|
| Voices    | 1-9    | 1       | Number of unison voices. 7 = authentic JP-8000 |
| Spread    | 0-100% | 10%     | Detune spread between voices                   |
| Mix       | 0-100% | 75%     | Balance between center and detuned voices      |

### Detune (Symmetric)

Adds symmetrically detuned copies of the signal for classic chorus-like thickening without the modulation. Useful for pads and leads.

| Parameter | Range       | Default | Description                          |
|-----------|-------------|---------|--------------------------------------|
| Voices    | 1-9         | 5       | Number of detune voices              |
| Amount    | 0-700 cents | 20 ct   | Detune spread in cents               |
| Mix       | 0-100%      | 0%      | Blend between dry and detuned signal |

## Effects

### Chorus

Four distinct chorus algorithms for adding width and movement:

- CE3 - Subtle, slow modulation inspired by Boss CE-3
- JP8 - JP-8000 style, optimized for Supersaw sounds
- AJ2-S / AJ2-M - Alpha Juno 2 Standard with opposite-phase LFOs
- AJ2-M - Alpha Juno 2 Matched with same-phase + polarity inversion

| Parameter | Range   | Default | Description                                                                                               |
|-----------|---------|---------|-----------------------------------------------------------------------------------------------------------|
| Mode      | 6 types | JP8     | Chorus algorithm: CE3, JP8, AJ2-S, AJ2-M, RCE10 (Boss DC-3 style ensemble), JP8-SC (JP-8000 Super Chorus) |



| Parameter          | Range   | Default | Description                                                                            |
|--------------------|---------|---------|----------------------------------------------------------------------------------------|
| Rate               | 0-100%  | 50%     | Modulation speed                                                                       |
| Depth              | 0-100%  | 100%    | Modulation intensity                                                                   |
| Mix                | 0-100%  | 0%      | Wet/dry balance                                                                        |
| Tone               | 0-100%  | 50%     | Brightness of chorus signal                                                            |
| Pre-Delay (Chorus) | 0-35 ms | 0 ms    | Delay ahead of the modulated tap, for swirl and doubling.<br>Works with all six modes. |

## Delay

Stereo delay with tempo sync, tone shaping and switchable Ping-Pong / Stereo modes.

| Parameter                  | Range              | Default   | Description                                                                      |
|----------------------------|--------------------|-----------|----------------------------------------------------------------------------------|
| Time                       | 1-2000 ms          | 375 ms    | Delay time (when not synced)                                                     |
| Delay Sync (Sync Division) | 24 values          | 1/2D      | Tempo-synced delay time                                                          |
| Feedback                   | 0-100%             | 50%       | Amount of signal fed back                                                        |
| Tone                       | 500-15000 Hz       | 3750 Hz   | Lowpass filter on delay signal                                                   |
| Mix                        | 0-100%             | 0%        | Wet/dry balance                                                                  |
| HP (Delay HP)              | 10-1000 Hz         | 10 Hz     | High-pass on the repeats, so long tails stay out of the low end.<br>10 Hz = off. |
| Mode (Delay Mode)          | Ping-Pong / Stereo | Ping-Pong | Ping-Pong cross-feeds L/R; Stereo keeps the channels independent.                |

## Reverb

Lush algorithmic reverb with pre-delay and sidechain ducking.

| Parameter      | Range      | Default | Description                                                                      |
|----------------|------------|---------|----------------------------------------------------------------------------------|
| Pre-Delay      | 0-250 ms   | 125 ms  | Initial delay before reverb onset                                                |
| Decay          | 0.01-20 s  | 6 s     | Reverb tail length                                                               |
| Damping        | 0-100%     | 50%     | High frequency absorption                                                        |
| Mix            | 0-100%     | 0%      | Wet/dry balance                                                                  |
| Duck Amount    | 0-100%     | 0%      | Sidechain compression from dry signal                                            |
| Duck Time      | 10-500 ms  | 100 ms  | Ducking recovery time                                                            |
| HP (Reverb HP) | 10-1000 Hz | 10 Hz   | High-pass on the reverb input, so lows stay tight. 10 Hz = off.                  |
| Pre-Delay Sync | Off / On   | Off     | Locks reverb pre-delay to host tempo, using the same divisions as LFO and Delay. |

## Arpeggiator

The SH-101 inspired arpeggiator offers 14 patterns including octave modes and wrapped polyrhythms for complex sequences.

### Arpeggio Patterns

| Pattern      | Description                              |
|--------------|------------------------------------------|
| Off          | Arpeggiator disabled                     |
| Up           | Ascending through held notes (1 octave)  |
| Down         | Descending through held notes (1 octave) |
| Up&Down      | Ascending then descending (1 octave)     |
| Down&Up      | Descending then ascending (1 octave)     |
| Up 2oct      | Ascending through 2 octaves              |
| Down 2oct    | Descending through 2 octaves             |
| Up&Down 2oct | Up then down through 2 octaves           |
| Down&Up 2oct | Down then up through 2 octaves           |
| Up W3        | Ascending with wrap every 3 steps        |
| Up W4        | Ascending with wrap every 4 steps        |
| Down W4      | Descending with wrap every 4 steps       |
| Up&Down W5   | Up/down with wrap every 5 steps          |
| Down 2oct W6 | Down 2 octaves with wrap every 6 steps   |

### Parameters

| Parameter | Range       | Default | Description                                                          |
|-----------|-------------|---------|----------------------------------------------------------------------|
| Type      | 14 patterns | Off     | Arpeggio pattern selection                                           |
| Step      | 9 divisions | 1/16    | Note division (1/4, 1/4T, 1/4D, 1/8, 1/8T, 1/8D, 1/16, 1/16T, 1/16D) |
| Gate      | 1-100%      | 80%     | Note length as percentage of step                                    |

## Performance Controls

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### Voice Mode

| Parameter            | Range         | Default | Description                                                                                                          |
|----------------------|---------------|---------|----------------------------------------------------------------------------------------------------------------------|
| V Mode (Voice Mode)  | Poly/Mono     | Poly    | Polyphonic or monophonic operation                                                                                   |
| Portamento           | 0-2 s         | 0 s     | Glide duration between notes. Constant-time: a semitone and an octave take the same time.                            |
| P Mode (Porta Mode)  | Always/Legato | Always  | Legato: glide only on overlapping notes                                                                              |
| Analog (Analog Feel) | 0-100%        | 0%      | Per-voice variance in tuning, cutoff, level and envelope timing. 5-15% is the classic zone; 0% is perfectly digital. |

### Stereo & Tone

| Parameter            | Range           | Default | Description                                 |
|----------------------|-----------------|---------|---------------------------------------------|
| Spread (Pan Spread)  | 0-100%          | 0%      | Random pan offset per voice                 |
| Width (Stereo Width) | 0-200%          | 100%    | Stereo field width (0=mono, 200=extra wide) |
| Bass                 | -15 to +15 dB   | 0 dB    | Low shelf EQ at 120 Hz                      |
| Mid                  | -15 to +15 dB   | 0 dB    | Peak EQ at 1.5 kHz                          |
| Mid Freq             | -300 Hz-2700 Hz | 1500 Hz | Sweepable mid frequency                     |
| Treble               | -15 to +15 dB   | 0 dB    | High shelf EQ at 3.5 kHz                    |
| OTT                  | 0-100%          | 0       | Effect depth/intensity                      |
| OTT Position         | Post/Pre        | Post    | Apply after or before reverb                |

## OTT

Multiband dynamics processor inspired by Ableton's OTT preset. Applies parallel upward and downward compression across three frequency bands for punchy, in-your-face dynamics.

### Output

| Parameter                | Range           | Default | Description                         |
|--------------------------|-----------------|---------|-------------------------------------|
| Level (Master)           | -60 to +12 dB   | 0 dB    | Master output level                 |
| PB Range                 | 0-24 st         | 12 st   | Pitch bend wheel range in semitones |
| Trsp (Transpose)         | -24 to +24 st   | 0 st    | Global pitch offset                 |
| Fine (Fine Tune)         | -100 to +100 ct | 0 ct    | Fine pitch adjustment in cents      |
| Vel Sens (Velocity Sens) | 0-100%          | 0%      | Velocity to volume sensitivity      |

# Installation

## System Requirements

- Windows 10/11 (64-bit) or macOS 10.13+
- VST3 or AU compatible DAW
- 8 GB RAM minimum
- 500 MB disk space

## Plugin Formats

RaveStation 2 is available in VST3 and AU formats. If you are using Logic Pro, install the AU version. For all other DAWs, use the VST3 version.



## Support

For technical support, updates, and additional information:

**Website:** [ravegeneration.io](http://ravegeneration.io)

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